Binary Classification using Logistic Regression

Predicting Customer Churn on the Telco Customer Churn Dataset

Technical Project Report

1. Executive Summary

This report documents an end-to-end machine learning workflow to predict customer churn for a telecommunications provider using Logistic Regression. Working from the Telco Customer Churn dataset (7,043 customers, 20 features), the project covers exploratory data analysis, data preprocessing, model training, evaluation, and interpretation of results, concluding with business-relevant insights and recommendations for future improvement.

The final model achieves a ROC-AUC of 0.84 and correctly identifies 78% of customers who actually churn (recall), giving the business a practical early-warning tool for targeting retention efforts.

2. Problem Definition

Customer churn — a subscriber ending their relationship with the service provider — directly reduces recurring revenue. Acquiring a new customer is typically far more expensive than retaining an existing one, so the ability to flag at-risk customers in advance, and to understand which factors drive their decision to leave, allows the business to intervene with targeted retention offers before losing them.

This is framed as a binary classification problem: given a customer's account and service attributes, predict whether they will churn (Yes) or not (No).

3. Dataset

The Telco Customer Churn dataset is a well-known IBM sample dataset distributed publicly on Kaggle. It contains 7,043 customer records and 21 columns, including demographic attributes, account information, subscribed services, and the churn label.

Attribute	Value
Total records	7,043
Total features	20 (excluding customer ID and target)
Numerical features	tenure, MonthlyCharges, TotalCharges
Categorical features	17 (gender, contract type, service subscriptions, payment method, etc.)
Target variable	Churn (Yes / No)
Class balance	73.5% No Churn / 26.5% Churn

4. Exploratory Data Analysis

4.1 Target Class Distribution

The target variable is imbalanced: roughly 73% of customers did not churn versus 27% who did. This imbalance was carried through into model training decisions (see Section 6).

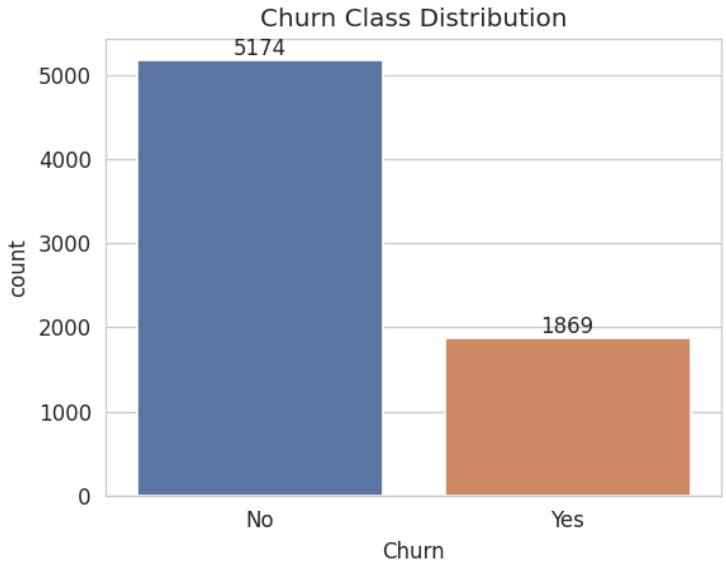


Figure 1: Distribution of the Churn target variable.

4.2 Numerical Feature Distributions

Tenure is heavily right-skewed with concentrations at both very low and very high values, suggesting distinct groups of brand-new and long-standing customers. MonthlyCharges shows a bimodal pattern, and TotalCharges is right-skewed as expected given its dependence on tenure.

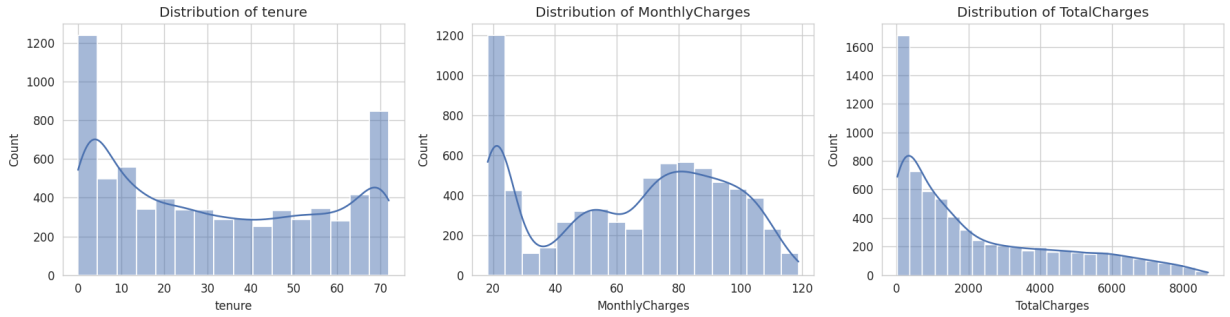


Figure 2: Distributions of tenure, MonthlyCharges, and TotalCharges.

4.3 Numerical Features vs. Churn

Customers who churned tend to have noticeably lower tenure and higher MonthlyCharges than those who stayed, an early signal that newer, higher-paying customers are more likely to leave.

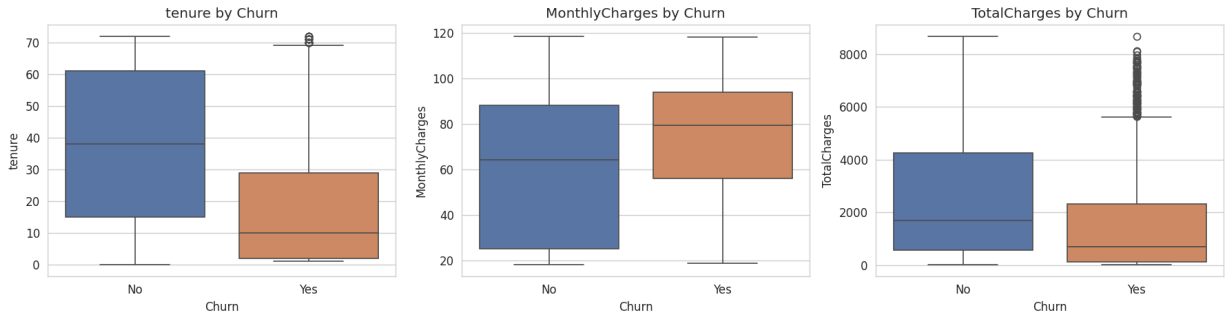


Figure 3: Boxplots of numerical features split by churn outcome.

4.4 Correlation Analysis

Tenure and TotalCharges are strongly positively correlated with each other (as expected) and both are negatively correlated with churn, while MonthlyCharges is positively correlated with churn.

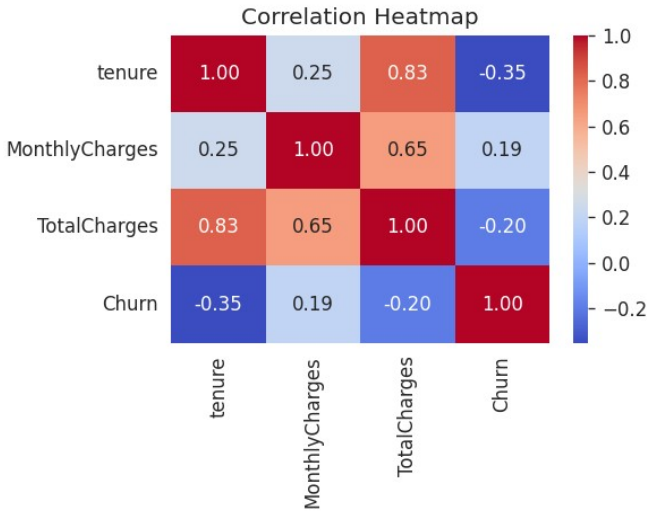


Figure 4: Correlation heatmap of numerical features and churn.

4.5 Churn Rate by Category

Breaking churn rate down by key categorical features reveals that month-to-month contracts, Fiber optic internet service, and electronic check payment are all associated with substantially higher churn rates than their alternatives.

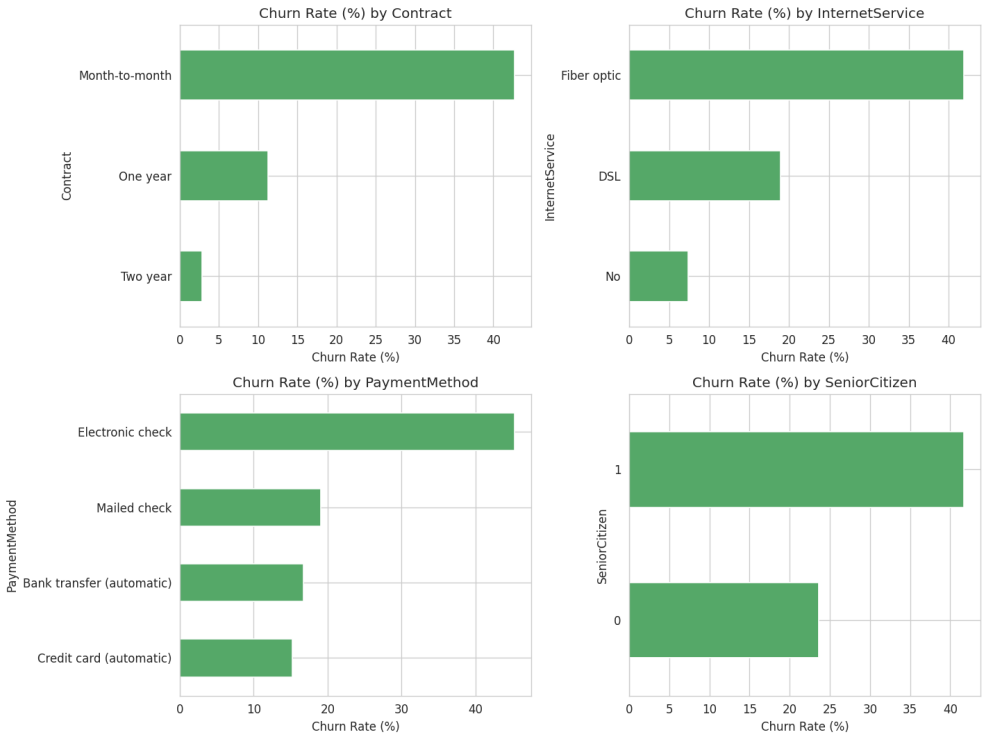


Figure 5: Churn rate (%) by Contract, InternetService, PaymentMethod, and SeniorCitizen status.

5. Data Preprocessing

5.1 Missing Values

The TotalCharges column contained 11 blank entries, all corresponding to customers with tenure = 0 (brand-new customers not yet billed). These were imputed with 0, which reflects reality rather than fabricating a history via mean imputation.

5.2 Encoding

The target variable was encoded to 1 (Yes) / 0 (No). All 16 categorical features were one-hot encoded using pandas.get_dummies with drop_first=True to avoid the dummy variable trap, expanding the feature space from 19 raw predictors to 30 encoded features.

5.3 Feature Scaling

The three numerical features (tenure, MonthlyCharges, TotalCharges) were standardized (zero mean, unit variance) using scikit-learn's StandardScaler. The scaler was fit only on the training split and then applied to the test split, preventing data leakage.

5.4 Train-Test Split

Data was split 80/20 into training and test sets using stratified sampling on the target variable, preserving the 73.5%/26.5% class ratio in both splits (5,634 training rows, 1,409 test rows).

6. Model Implementation

A Logistic Regression classifier (scikit-learn) was trained with the following hyperparameters:

Hyperparameter	Value / Justification
penalty	l2 — standard ridge-style regularization, keeps all one-hot encoded features while shrinking noisy coefficients
C	1.0 — moderate regularization strength; default value was appropriate given ~30 features vs. ~5,600 training rows (no strong over/underfitting signal)
solver	lbfgs — efficient default solver for L2-penalized logistic regression at this dataset scale
class_weight	balanced — reweights the loss inversely to class frequency to counteract the 73/27 imbalance
max_iter	1000 — increased from default to ensure convergence

7. Model Evaluation

The trained model was evaluated on the held-out 20% test set (1,409 customers).

Metric	Value
Accuracy	0.7381
Precision (Churn)	0.5043
Recall (Churn)	0.7834
F1 Score (Churn)	0.6136
ROC-AUC	0.8417

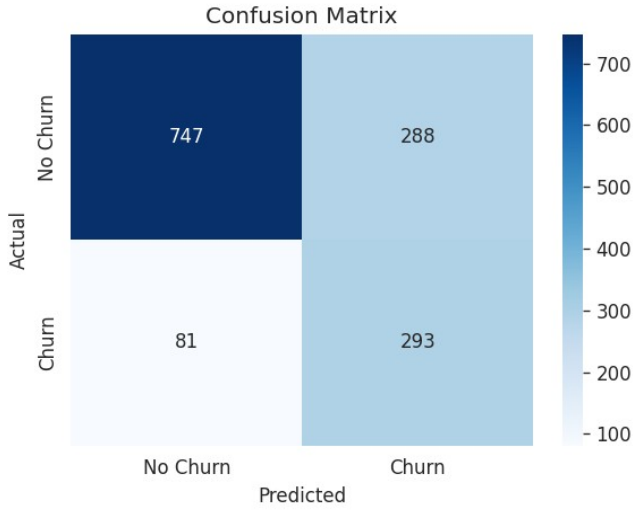


Figure 6: Confusion matrix on the test set (747 true negatives, 288 false positives, 81 false negatives, 293 true positives).

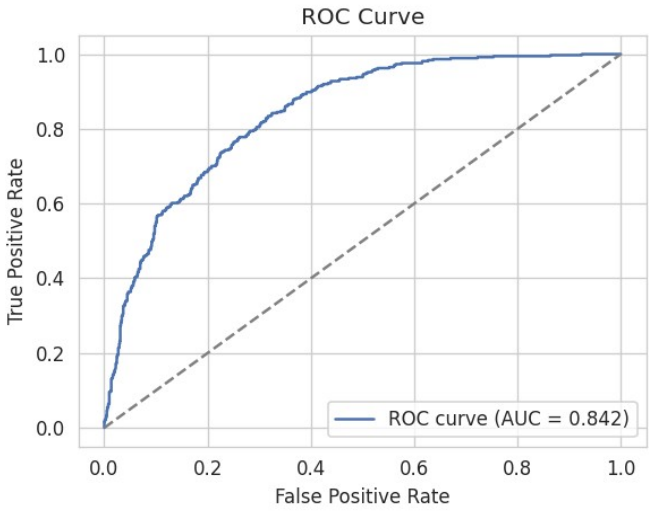


Figure 7: ROC curve, AUC = 0.842.

Accuracy alone is a weak summary metric here because of class imbalance — a naive model that always predicts "No Churn" would already score ~73.5% accuracy. The class_weight='balanced' setting deliberately trades some precision for higher recall on the Churn class: catching 78% of customers who actually churn is prioritized over minimizing false alarms, since the cost of a missed churner is typically higher than the cost of an unnecessary retention offer. The ROC-AUC of 0.84 indicates the model has strong overall discriminative ability independent of the classification threshold.

8. Interpretation & Insights

Because Logistic Regression coefficients act directly on the log-odds of churn, they are directly interpretable: a positive coefficient increases the predicted probability of churn, and a negative coefficient decreases it.

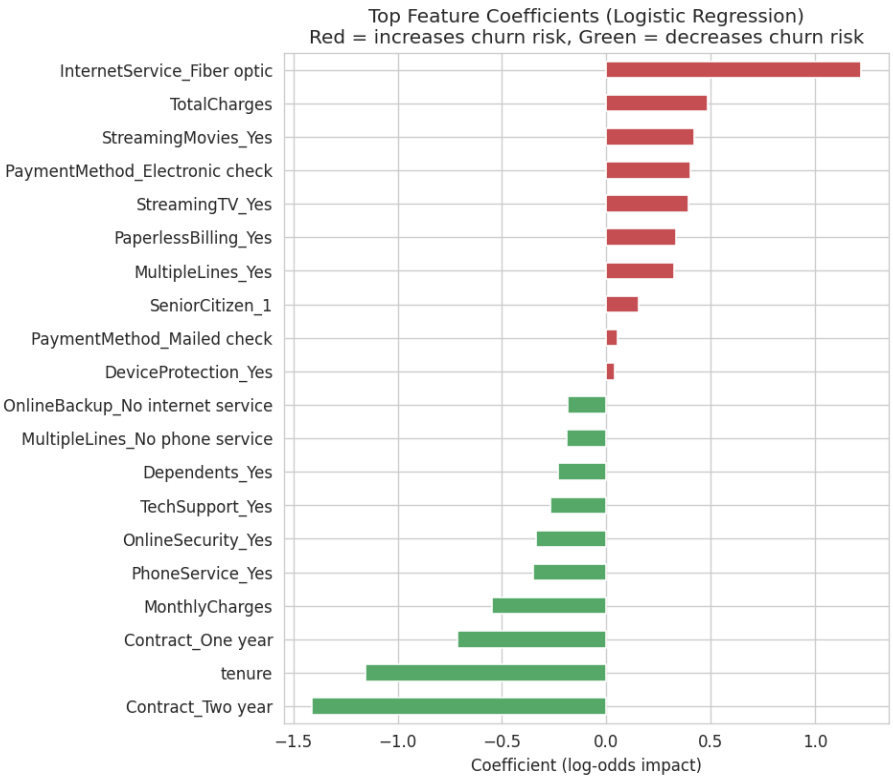


Figure 8: Top 10 churn-increasing and top 10 churn-decreasing feature coefficients.

8.1 Top Churn-Increasing Drivers

Feature	Coefficient
InternetService: Fiber optic	+1.222
TotalCharges	+0.487
StreamingMovies: Yes	+0.421
PaymentMethod: Electronic check	+0.402
StreamingTV: Yes	+0.395
PaperlessBilling: Yes	+0.336
MultipleLines: Yes	+0.325

8.2 Top Churn-Decreasing Drivers

Feature	Coefficient
Contract: Two year	-1.413
tenure	-1.156
Contract: One year	-0.715
MonthlyCharges	-0.546
OnlineSecurity: Yes	-0.335
TechSupport: Yes	-0.269
Dependents: Yes	-0.231

Contract length is by far the strongest churn-reducing factor: customers on two-year or one-year contracts are far less likely to leave than those on month-to-month agreements (the implicit baseline). Longer tenure similarly reduces churn risk, consistent with the EDA. On the risk-increasing side, Fiber optic internet stands out as the single

largest positive coefficient — despite being a premium service, it correlates with higher churn, possibly reflecting price sensitivity or service quality concerns. Add-on services like online security and tech support reduce churn, suggesting that customers more "invested" in the service ecosystem are more likely to stay.

9. Conclusion & Future Work

9.1 Key Findings

- A Logistic Regression model trained on standard account and service attributes achieves 0.84 ROC-AUC and 78% recall on churners, providing a practical early-warning signal.
- Contract length and tenure are the strongest churn-reducing signals; Fiber optic internet, month-to-month contracts, and electronic check payment are the strongest churn-increasing signals.
- These findings map directly onto retention strategies: incentivizing longer contract commitments, auditing Fiber optic service satisfaction, and proactively engaging early-tenure customers.

9.2 Limitations

- The dataset is a single static snapshot and does not capture how customer behavior changes over time.
- Logistic Regression assumes a linear relationship with the log-odds of churn and cannot model complex non-linear feature interactions.
- class_weight='balanced' improves recall at the cost of precision; the ideal trade-off depends on real business costs for false positives vs. false negatives, which were not available for this project.
- External factors such as competitor pricing, customer service interactions, and complaints are not captured in the dataset.

9.3 Future Work

- Apply SMOTE (Synthetic Minority Oversampling) to the training data and compare against the class_weight='balanced' approach used here.
- Evaluate more expressive models such as Random Forest or XGBoost, which can capture non-linear feature interactions.
- Engineer additional features, e.g. TotalCharges / tenure as an effective average monthly spend, or interaction terms between Contract and tenure.
- Tune the classification threshold against a business-defined cost matrix rather than using the default 0.5 cutoff.
- Deploy the model as a live scoring service (see Section 10) so the business can score its current customer base on a recurring basis.

10. Deployment

The trained model, scaler, and feature schema were serialized to logistic_regression_model.pkl. A minimal Streamlit application (app/streamlit_app.py) was built on top of this artifact, allowing a user to enter a customer's attributes and receive a live churn-risk probability. The application can be run locally with 'streamlit run streamlit_app.py', or deployed for free on Streamlit Community Cloud, or containerized with Docker for deployment to any cloud platform (AWS, GCP, Azure, Render, etc.). Full setup instructions are provided in the accompanying chat walkthrough and README.md.

11. Submission Package Contents

- Report.pdf — this document
- Churn_Prediction_Logistic_Regression.ipynb — fully executed Jupyter notebook

- Dataset.csv / Dataset_clean.csv — raw and cleaned data
- charts/ — exported PNG visualizations
- README.md — project documentation
- logistic_regression_model.pkl, churn_logistic_regression.py, app/streamlit_app.py — supporting code and deployable artifacts